UNIT-1

Database: Database is a collection of inter-related data which helps in efficient retrieval, insertion and deletion of data from database and organizes the data in the form of tables, views, schemas, reports etc. For Example, university database organizes the data about students, faculty, and admin staff etc. which helps in efficient retrieval, insertion and deletion of data from it.

DBMS stands for **D**ata**b**ase **M**anagement **S**ystem. We can break it like this DBMS = Database + Management System. Database is a collection of data and Management System is a set of programs to store and retrieve those data. Based on this we can **define DBMS** like this: DBMS is a collection of inter-related data and set of programs to store & access those data in an easy and effective manner.

Need of DBMS:- Database systems are basically developed for large amount of data. When dealing with huge amount of data, there are two things that require optimization: **Storage of data** and **retrieval of data**.

DBMS Vs File system

Drawbacks of File system

- **Data redundancy:** Data redundancy refers to the duplication of data, lets say we are managing the data of a college where a student is enrolled for two courses, the same student details in such case will be stored twice, which will take more storage than needed. Data redundancy often leads to higher storage costs and poor access time.
- **Data inconsistency:** Data redundancy leads to data inconsistency, lets take the same example that we have taken above, a student is enrolled for two courses and we have student address stored twice, now lets say student requests to change his address, if the address is changed at one place and not on all the records then this can lead to data inconsistency.
- **Data Isolation:** Because data are scattered in various files, and files may be in different formats, writing new application programs to retrieve the appropriate data is difficult.
- **Dependency on application programs:** Changing files would lead to change in application programs.
- **Atomicity issues:** Atomicity of a transaction refers to “All or nothing”, which means either all the operations in a transaction executes or none.
- **Data Security:** Data should be secured from unauthorised access, for example a student in a college should not be able to see the payroll details of the teachers, such kind of security constraints are difficult to apply in file processing systems.

Advantage of DBMS over file system

There are several advantages of Database management system over file system. Few of them are as follows:

- **No redundant data:** Redundancy removed by data normalization. No data duplication saves storage and improves access time.

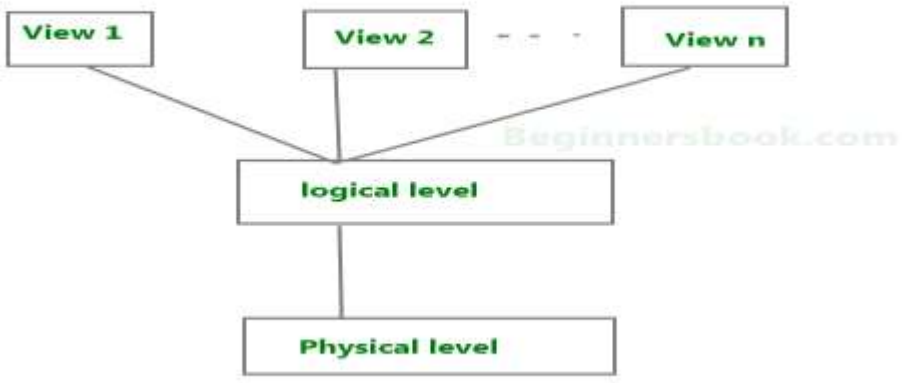
- **Data Consistency and Integrity:** As we discussed earlier the root cause of data inconsistency is data redundancy, since data normalization takes care of the data redundancy, data inconsistency also been taken care of as part of it
- **Data Security:** It is easier to apply access constraints in database systems so that only authorized user is able to access the data. Each user has a different set of access thus data is secured from the issues such as identity theft, data leaks and misuse of data.
- **Privacy:** Limited access means privacy of data.
- **Easy access to data**:- Database systems manages data in such a way so that the data is easily accessible with fast response times.
- **Easy recovery:** Since database systems keeps the backup of data, it is easier to do a full recovery of data in case of a failure.
- **Flexible:** Database systems are more flexible than file processing systems.

Disadvantages of DBMS:

- DBMS implementation cost is high compared to the file system
- Complexity: Database systems are complex to understand
- Performance: Database systems are generic, making them suitable for various applications. However this feature affect their performance for some applications

Data Abstraction in DBMS

Database systems are made-up of complex data structures. To ease the user interaction with database, the developers hide internal irrelevant details from users. This process of hiding irrelevant details from user is called data abstraction.



Three Levels of data abstraction

We have three levels of abstraction:

Physical level: This is the lowest level of data abstraction. It describes how data is actually stored in database. You can get the complex data structure details at this level.

Logical level: This is the middle level of 3-level data abstraction architecture. It describes what data is stored in database.

View level: Highest level of data abstraction. This level describes the user interaction with database system.

Physical Level:

- Lowest level of abstraction.
- Describes how the data are stored.
- Complex low-level data structures described in detail.
- E.g: index, B-tree, hashing.

Logical / Conceptual Level:

- Next highest level of abstraction.
- Describes what data are stored and what relationships exit among those data.
- Database administrator level.

View Level:

- Highest level of abstraction.
- Describes only part of the database for a particular group of users.
- Can be many different views of a database.

Data Independence

A database system normally contains a lot of data in addition to users' data. For example, it stores data about data, known as metadata, to locate and retrieve data easily. It is rather difficult to modify or update a set of metadata once it is stored in the database.

Types of Data Independence

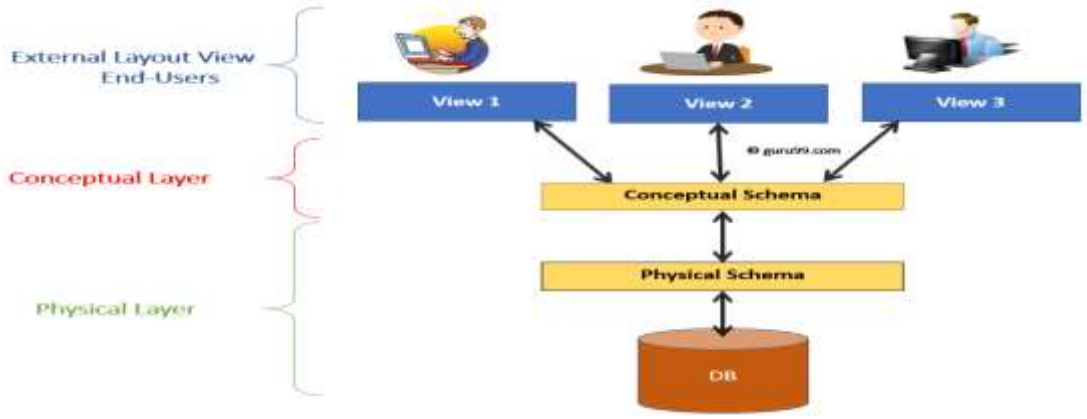
In DBMS there are two types of data independence

1. Physical data independence
2. Logical data independence.

Levels of Database

Before we learn Data Independence, a refresher on Database Levels is important. The database has 3 levels as shown in the diagram below

1. Physical/Internal
2. Conceptual
3. External



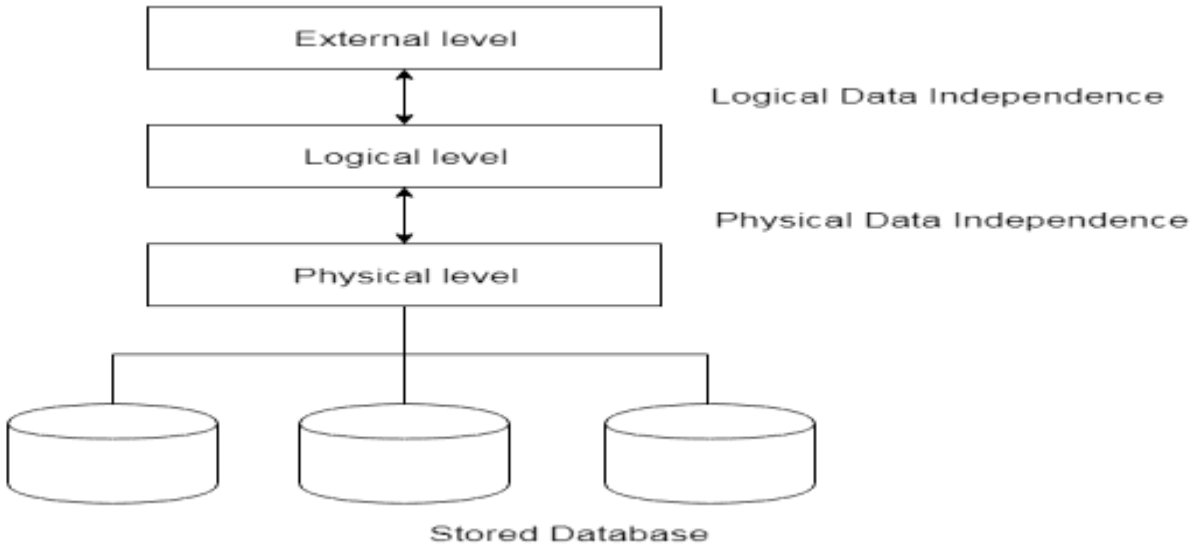
Levels of DBMS Architecture Diagram

1.Logical Data Independence

- Logical data independence refers characteristic of being able to change the conceptual schema without having to change the external schema.
- Logical data independence is used to separate the external level from the conceptual view.
- If we do any changes in the conceptual view of the data, then the user view of the data would not be affected.
- Logical data independence occurs at the user interface level.

2. Physical Data Independence

- Physical data independence can be defined as the capacity to change the internal schema without having to change the conceptual schema.
- If we do any changes in the storage size of the database system server, then the Conceptual structure of the database will not be affected.
- Physical data independence is used to separate conceptual levels from the internal levels.
- Physical data independence occurs at the logical interface level.



DBMS Architecture

Database management systems architecture will help us understand the components of database system and the relation among them.

The architecture of DBMS depends on the computer system on which it runs. For example, in a client-server DBMS architecture, the database systems at server machine can run several requests made by client machine. We will understand this communication with the help of diagrams.

Types of DBMS Architecture

There are three types of DBMS architecture:

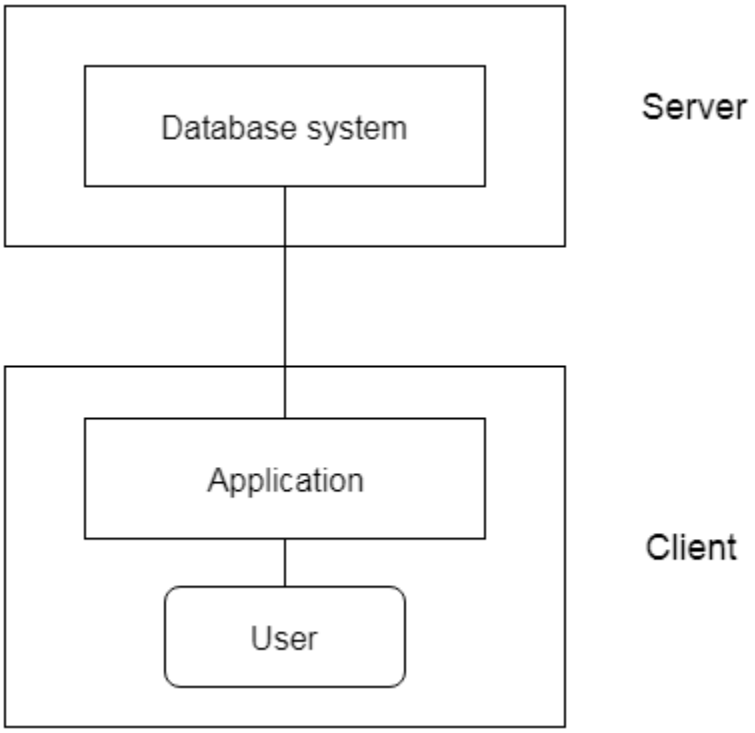
1. Single tier architecture
2. Two tier architecture
3. Three tier architecture

1. Single tier architecture

In this type of architecture, the database is readily available on the client machine, any request made by client doesn't require a network connection to perform the action on the database.

For example, lets say you want to fetch the records of employee from the database and the database is available on your computer system, so the request to fetch employee details will be done by your computer and the records will be fetched from the database by your computer as well. This type of system is generally referred as local database system.

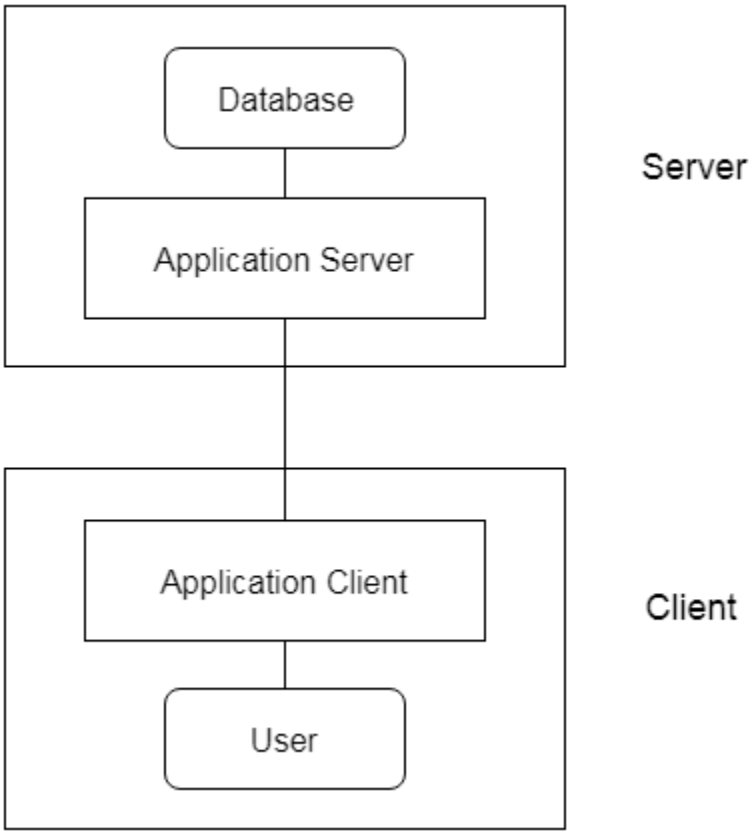
2. Two tier architecture



In two-tier architecture, the Database system is present at the server machine and the DBMS application is present at the client machine, these two machines are connected with each other through a reliable network as shown in the above diagram.

Whenever client machine makes a request to access the database present at server using a query language like sql, the server perform the request on the database and returns the result back to the client. The application connection interface such as JDBC, ODBC are used for the interaction between server and client.

3. Three tier architecture



In three-tier architecture, another layer is present between the client machine and server machine. In this architecture, the client application doesn't communicate directly with the database systems present at the server machine, rather the client application communicates with server application and the server application internally communicates with the database system present at the server.

DBMS Interfaces

A database management system (DBMS) interface is a user interface which allows for the ability to input queries to a database without using the query language itself. A DBMS interface could be a web client, a local client that runs on a desktop computer, or even a mobile app. User-friendly interfaces provided by a DBMS may include the following:

Menu-Based Interfaces for Web Clients or Browsing:- These interfaces pre-sent the user with lists of options (called menus) that lead the user through the formulation of a request. Menus do away with the need to memorize the specific commands and syntax of a query language; rather, the query is composed step-by-step by picking options from a menu that is displayed by the system. Pull-down menus are a very popular technique in Web-based user interfaces. They are also often used in browsing interfaces, which allow a user to look through the contents of a database in an exploratory and unstructured manner.

Forms-Based Interfaces. A forms-based interface displays a form to each user. Users can fill out all of the form entries to insert new data, or they can fill out only certain entries, in which case the DBMS will retrieve matching data for the remaining entries. Forms are usually designed and programmed for naive users as inter-faces to canned transactions. Many DBMSs have forms specification languages, which are special languages that help programmers specify such forms. SQL*Forms is a form-based language that specifies queries using a form designed in conjunction with the relational database schema. Oracle Forms is a component of the Oracle product suite that provides an extensive set of features to design and build applications using forms. Some systems have utilities that define a form by letting the end user interactively construct a sample form on the screen.

Graphical User Interfaces. A GUI typically displays a schema to the user in diagrammatic form. The user then can specify a query by manipulating the diagram. In many cases, GUIs utilize both menus and forms. Most GUIs use a pointing device, such as a mouse, to select certain parts of the displayed schema diagram.

Natural Language Interfaces. These interfaces accept requests written in English or some other language and attempt to understand them. A natural language interface usually has its own schema, which is similar to the database conceptual schema, as well as a dictionary of important words. The natural language interface refers to the words in its schema, as well as to the set of standard words in its dictionary, to interpret the request. If the interpretation is successful, the inter-face generates a high-level query corresponding to the natural language request and submits it to the DBMS for processing; otherwise, a dialogue is started with the user to clarify the request. The capabilities of natural language interfaces have not advanced rapidly. Today, we see search engines that accept strings of natural language (like English or Spanish) words and match them with documents at specific sites (for local search engines) or Web pages on the Web at large (for engines like Google or Ask). They use predefined indexes on words and use ranking functions to retrieve and present resulting documents in a decreasing degree of match. Such “free form” textual query interfaces are not yet common in structured relational or legacy model databases, although a research area called keyword-based querying has emerged recently for relational databases.

Speech Input and Output. Limited use of speech as an input query and speech as an answer to a question or result of a request is becoming commonplace. Applications with limited vocabularies such as inquiries for telephone directory, flight arrival/departure, and credit card account information are allowing speech for input and output to enable customers to access this information. The speech input is detected using a library of predefined words and used to set up the parameters that are supplied to the queries. For output, a similar conversion from text or numbers into speech takes place.

Interfaces for Parametric Users. Parametric users, such as bank tellers, often have a small set of operations that they must perform repeatedly. For example, a teller is able to use single function keys to invoke routine and repetitive transactions such as account deposits or withdrawals, or balance inquiries. Systems analysts and programmers design and implement a special interface for each known class of naive users. Usually a small set of abbreviated commands is included, with the goal of minimizing the number of keystrokes required for each request. For example, function keys in a terminal can be

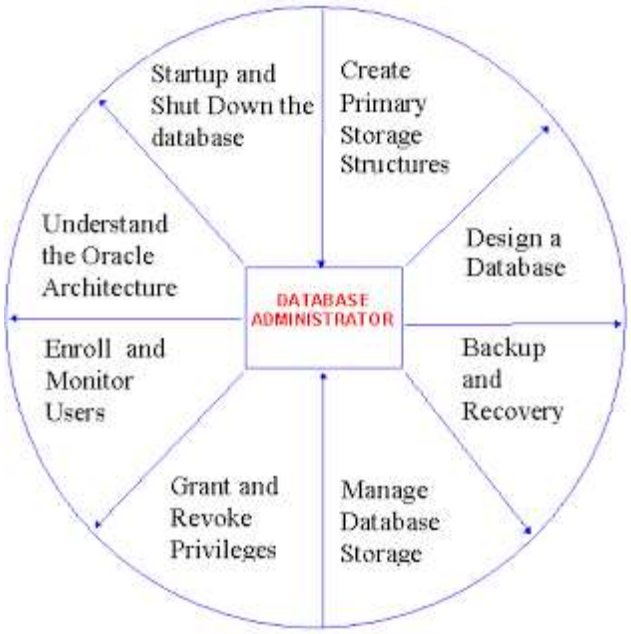
programmed to initiate various commands. This allows the parametric user to proceed with a minimal number of keystrokes.

Interfaces for the DBA. Most database systems contain privileged commands that can be used only by the DBA staff. These include commands for creating accounts, setting system parameters, granting account authorization, changing a schema, and reorganizing the storage structures of a database

Database User:- There are about 4 different types of database users differentiated by their way of interaction with the database system. Brief description has been given below,

1. **Naive Users:** This type of users generally interacts with the system through previously created programs. Such as, a user may use the system to transfer some balance from one account to another. This will be done a program, like 'Transfer'. The user will only fill some required fields of the program like the balance, the accounts etc. The typical interfaces for these users are forms including some required fields to be filled.
2. **Application Programmers:** Application programmers are computer professionals. They develop user interfaces. Rapid Application Development (RAD) is a tool that enables them to construct forms and reports with minimal programming effort.
3. **Sophisticated Users:** Sophisticated users form their requests in a database query language and submit them to a query processor to make the storage manager understand the requests. Analysts who submit queries to explore data in the database fall in this category.
4. **Specialized Users:** Specialized users are sophisticated users who write specialized database application that does not fit into the traditional data-processing framework. Among these applications are computer aided-design systems, knowledge-base and expert systems etc.

Definition of DBA:- Database administrator(DBA) is a professional Person or group of professional persons responsible for designing and maintaining the database.The role of database administrator is very important for successfull database execution.



Rolls Of DBA :

Some of the major roles of DBA of an organization listed below:-

1.Designing schemas :-

DBA designs the structure of complete database .The collection of schemas is decided by the DBA with the help of conceptual database definition.

2. Designing constants on database items:-

DBA designs the constant or rules of the data-items (attributes) which properly defined in the data dictionary of database manager. This contents ensure the integrity of the database, this keep the database in conscious state.

3.Modification to the physical schema structure:-

DBA is entitled to modify the physical structure when it is required or when some changes are made to the database.The modification is done using DDL commands are statements.

4.Authorisation of grants:-

The biggest goal of DBA is to decide what part of the database is accessed by whome. A user can only read the data and read or write the data or modify the data structure itself are few types of accesses, this kind of grant is the functionality of the DBA or decided by DBA.

5.Imposition of security:-

DBA has to control the data form unauthorised accesses, To improve the security of the data stored in the database Is decided by the DBA.In future routine maintenance is also done by the DBA In the case of some Special purpose.

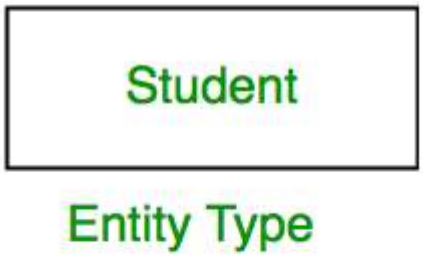
Introduction of ER Model

ER Model is used to model the logical view of the system from data perspective which consists of these components:

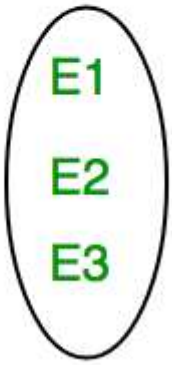
Entity
Entity Type
Entity Set

An Entity may be an object with a physical existence – a particular person, car, house, or employee – or it may be an object with a conceptual existence – a company, a job, or a university course.

An Entity is an object of Entity Type and set of all entities is called as entity set. e.g.; E1 is an entity having Entity Type Student and set of all students is called Entity Set. In ER diagram, Entity Type is represented as:



Entity Type

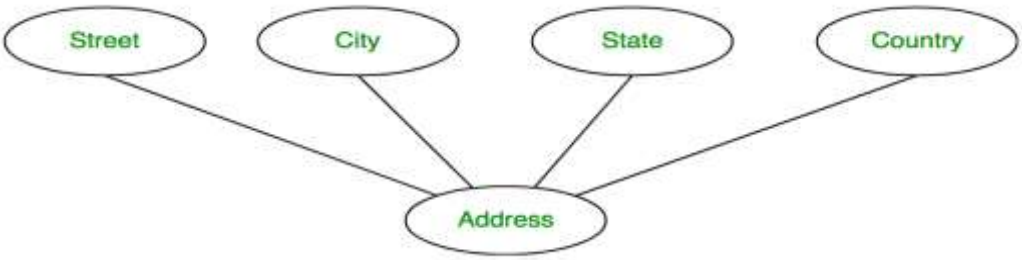


Entity Set

Attribute(s):
Attributes are the properties which define the entity type. For example, Roll_No, Name, DOB, Age, Address, Mobile_No are the attributes which defines entity type Student. In ER diagram, attribute is represented by an oval.

1. **Key Attribute –**
   The attribute which uniquely identifies each entity in the entity set is called key attribute.For example, Roll_No will be unique for each student. In ER diagram, key attribute is represented by an oval with underlying lines.

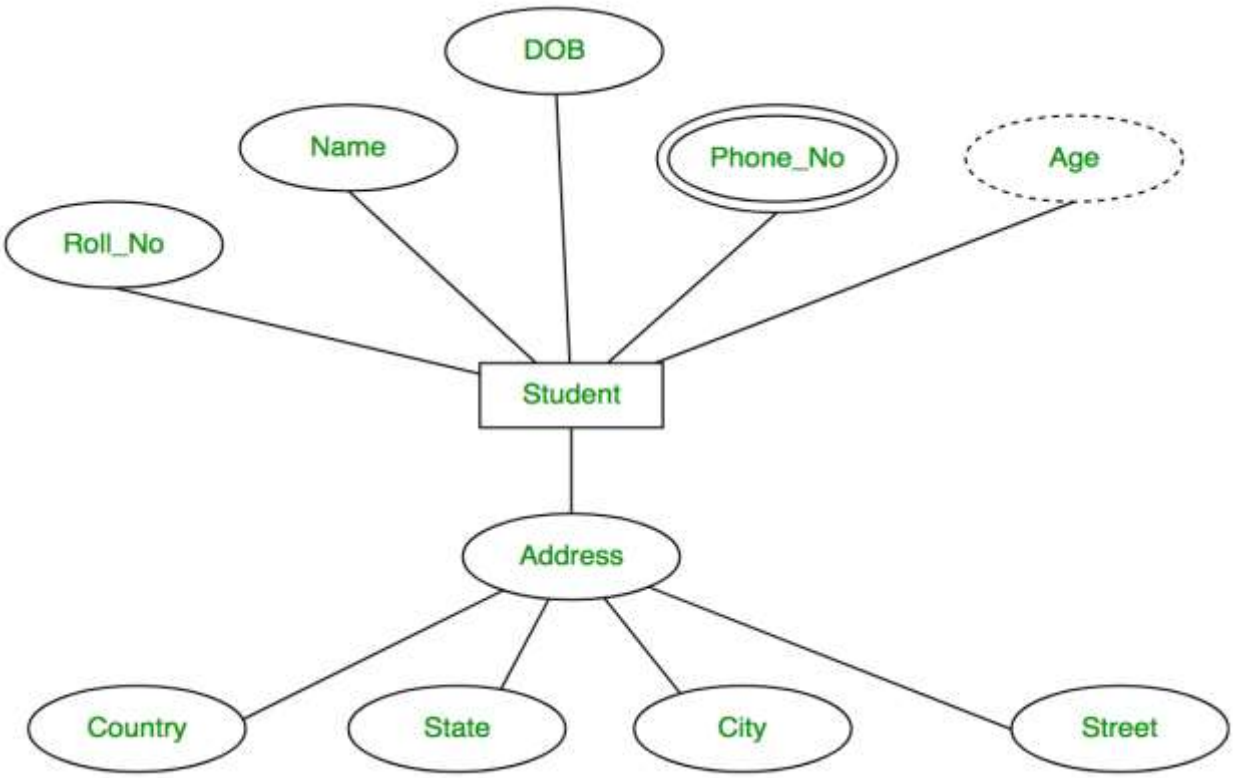
2. **Composite Attribute –**
   An attribute composed of many other attribute is called as composite attribute. For example, Address attribute of student Entity type consists of Street, City, State, and Country. In ER diagram, composite attribute is represented by an oval comprising of ovals.



3. **Multivalued Attribute –**
   An attribute consisting more than one value for a given entity. For example, Phone_No (can be more than one for a given student). In ER diagram, multivalued attribute is represented by double oval.

4. **Derived Attribute –**
   An attribute which can be derived from other attributes of the entity type is known as derived attribute. e.g.; Age (can be derived from DOB). In ER diagram, derived attribute is represented by dashed oval.

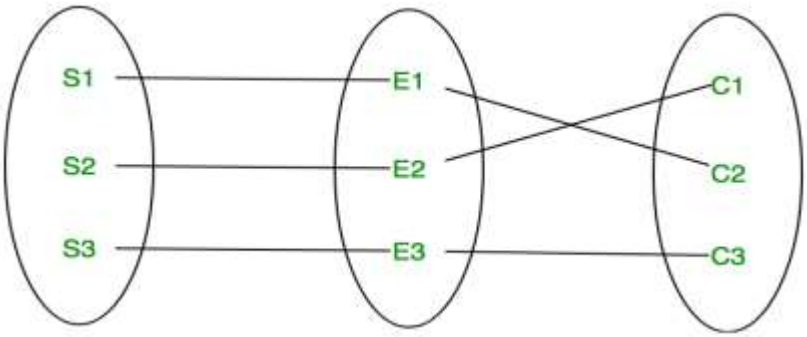
   The complete entity type Student with its attributes can be represented as:



Relationship Type and Relationship Set:

A relationship type represents the association between entity types. For example,'Enrolled in' is a relationship type that exists between entity type Student and Course. In ER diagram, relationship type is represented by a diamond and connecting the entities with lines.

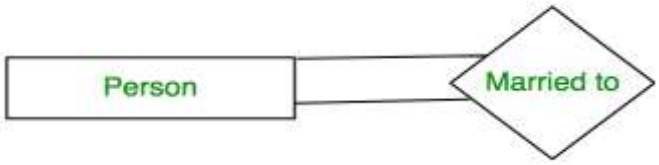
A set of relationships of same type is known as relationship set. The following relationship set depicts S1 is enrolled in C2, S2 is enrolled in C1 and S3 is enrolled in C3.



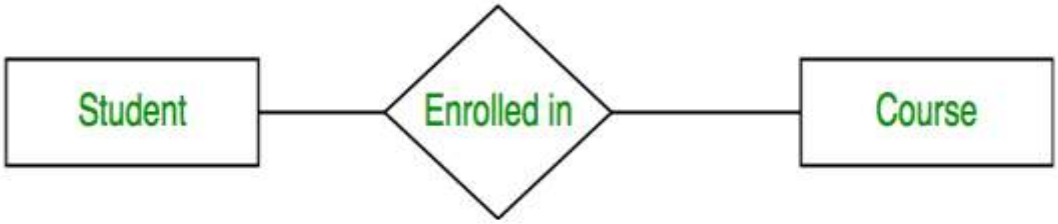
Degree of a relationship set:

The number of different entity sets participating in a relationship set is called as degree of a relationship set.

1. **<u>Unary Relationship –</u>**
   When there is only ONE entity set participating in a relation, the relationship is called as unary relationship. For example, one person is married to only one person.



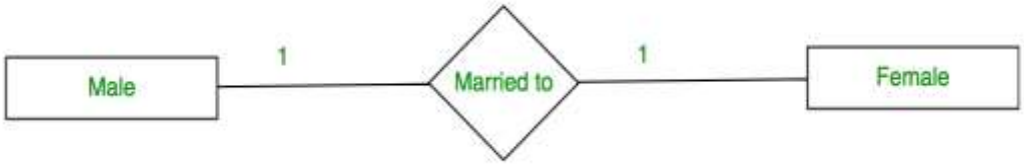
2. **<u>Binary Relationship –</u>**
   When there are TWO entities set participating in a relation, the relationship is called as binary relationship.For example, Student is enrolled in Course.



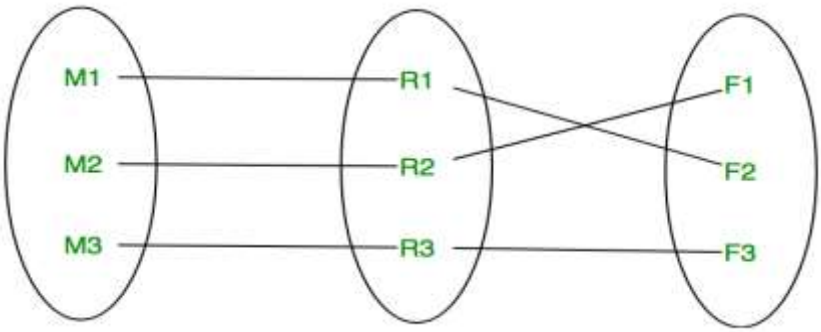
3. **<u>n-ary Relationship –</u>**
   When there are n entities set participating in a relation, the relationship is called as n-ary relationship.

<u>Cardinality:</u>
The number of times an entity of an entity set participates in a relationship set is known as cardinality. Cardinality can be of different types:

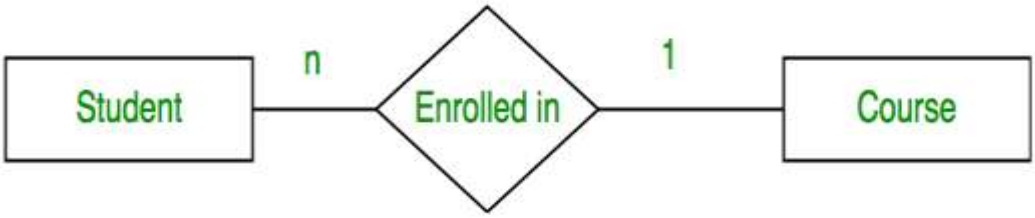
1. One to one – When each entity in each entity set can take part only once in the relationship, the cardinality is one to one. Let us assume that a male can marry to one female and a female can marry to one male. So the relationship will be one to one.



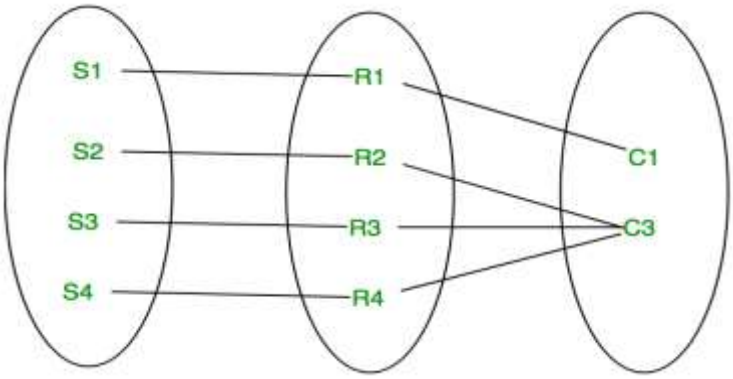
   Using Sets, it can be represented as:



2. Many to one – When entities in one entity set can take part only once in the relationship set and entities in other entity set can take part more than once in the relationship set, cardinality is many to one. Let us assume that a student can take only one course but one course can be taken by many students. So the cardinality will be n to 1. It means that for one course there can be n students but for one student, there will be only one course.



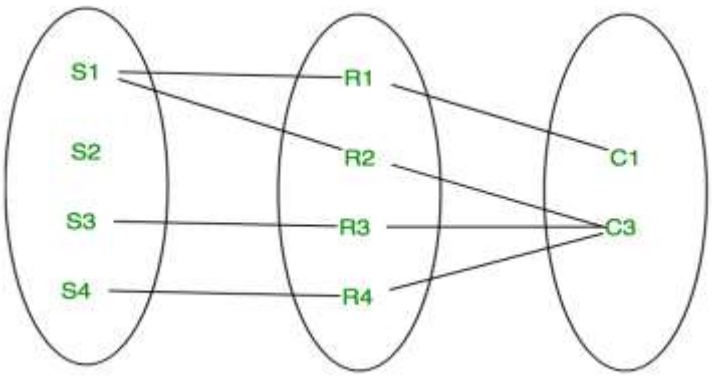
Using Sets, it can be represented as:



In this case, each student is taking only 1 course but 1 course has been taken by many students.

3. Many to many – When entities in all entity sets can take part more than once in the relationship cardinality is many to many. Let us assume that a student can take more than one course and one course can be taken by many students. So the relationship will be many to many.

Using sets, it can be represented as:



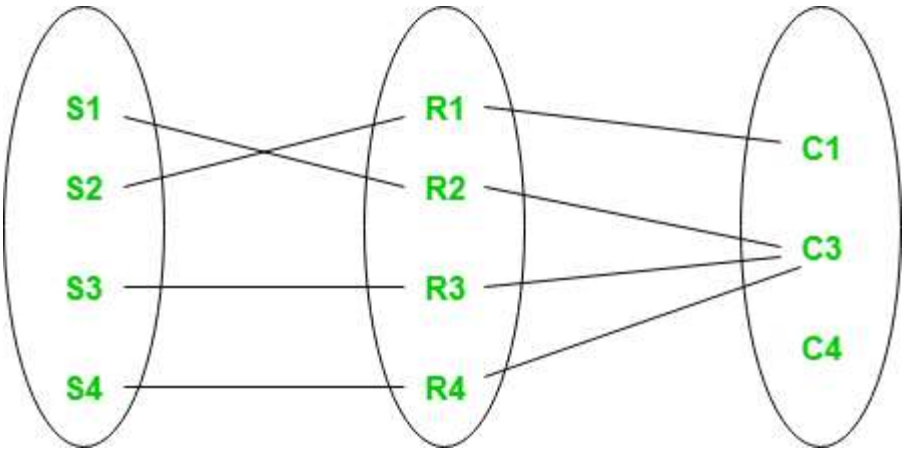
In this example, student S1 is enrolled in C1 and C3 and Course C3 is enrolled by S1, S3 and S4. So it is many to many relationships.

Participation Constraint:

Participation Constraint is applied on the entity participating in the relationship set.

1. Total Participation – Each entity in the entity set must participate in the relationship. If each student must enroll in a course, the participation of student will be total. Total participation is shown by double line in ER diagram.
2. Partial Participation – The entity in the entity set may or may NOT participate in the relationship. If some courses are not enrolled by any of the student, the participation of course will be partial. The diagram depicts the 'Enrolled in' relationship set with Student Entity set having total participation and Course Entity set having partial participation.

Using set, it can be represented as,



Every student in Student Entity set is participating in relationship but there exists a course C4 which is not taking part in the relationship.

Weak Entity Type and Identifying Relationship:

As discussed before, an entity type has a key attribute which uniquely identifies each entity in the entity set. But there exists some entity type for which key attribute can't be defined. These are called Weak Entity type.

For example, A company may store the information of dependants (Parents, Children, Spouse) of an Employee. But the dependents don't have existence without the employee. So Dependent will be weak entity type and Employee will be Identifying Entity type for Dependant.

A weak entity type is represented by a double rectangle. The participation of weak entity type is always total. The relationship between weak entity type and its identifying strong entity type is called identifying relationship and it is represented by double diamond.